Hierarchical Clustering is one of the easiest clustering algorithms to understand because it works very much like a **family tree**.

Imagine a Classroom of Students

Suppose we have 5 students and we know their heights:

Student	Height
A	150 cm
B	152 cm
C	155 cm
D	180 cm
E	182 cm

Goal: Find groups of similar students.

Agglomerative Clustering (Bottom-Up)

This is the most commonly used hierarchical clustering method.

Step 1: Everyone starts alone

```
A    B    C    D    E
```

Each student is its own cluster.

Step 2: Merge the closest pair

Who are closest?

- A (150) and B (152)
- Difference = 2 cm

Merge them:

```
(A,B)    C    D    E
```

Step 3: Find next closest clusters

Now compare:

- (A,B) with C
- D with E

D and E are only 2 cm apart.

Merge them:

```
(A,B)   C   (D,E)
```

Step 4: Merge again

C (155) is very close to cluster (A,B).

Merge:

```
(A,B,C)   (D,E)
```

Step 5: Final merge

Now only two clusters remain.

Merge them:

```
(A,B,C,D,E)
```

One giant cluster.

The Family Tree (Dendrogram)

The algorithm builds a tree like this:

```
           All Students
              /     \
             /       \
        (ABC)       (DE)
        /   \       /  \
      (AB)   C     D    E
      / \
     A   B
```

This tree is called a **Dendrogram**.

How Do We Decide Number of Clusters?

This is the biggest advantage over K-Means.

In K-Means:

❌ You must decide K beforehand.

Example:

```
K = 2
```

But what if the data naturally has 3 groups?

In Hierarchical Clustering:

✅ Build the whole tree first.

Then simply "cut" the tree where it makes sense.

Cut Here → 2 Clusters

```
           All Students
              /     \
             /------\
        (ABC)        (DE)
```

Result:

Cluster 1 = A,B,C

Cluster 2 = D,E

Cut Lower → 3 Clusters

```
        (ABC)        (DE)
        /    \
      (AB)    C
```

Result:

Cluster 1 = A,B

Cluster 2 = C

Cluster 3 = D,E

Real-Life Example: Online Shopping

Suppose Amazon has customers:

Customer	Purchases
John	Mobile, Earbuds
Mary	Mobile, Laptop
Tom	Vegetables
Sara	Fruits
Alex	Mobile, Tablet

Agglomerative clustering would:

1. Put everyone separately
2. Merge John + Mary
3. Merge Alex with them
4. Merge Tom + Sara
5. Build a tree

Final clusters:

```
Tech Buyers
 ├─ John
 ├─ Mary
 └─ Alex

Grocery Buyers
 ├─ Tom
 └─ Sara
```

Amazon can then target different advertisements to each cluster.

Divisive Clustering (Top-Down)

This is the opposite approach.

Start with:

```
(A,B,C,D,E)
```

One giant cluster.

Split into two groups:

```
(A,B,C)   (D,E)
```

Then split again:

```
(A,B)   C   (D,E)
```

And so on.

Think of it like:

🏢 A company starts as one team

→ Sales + Engineering

→ Engineering splits into Backend + Frontend

→ Backend splits into API + Database teams

This creates the same type of tree but from the top.

K-Means vs Hierarchical Clustering

Feature	K-Means	Hierarchical
Need K beforehand?	Yes	No
Creates tree?	No	Yes
Easy to visualize?	Medium	Very Easy
Works well on small datasets?	Good	Excellent
Works well on huge datasets?	Excellent	Slower

Simple Memory Trick

👪 **Hierarchical Clustering = Building a Family Tree**

- **Agglomerative** = Children join to form families → families join to form bigger families.
- **Divisive** = One big family splits into smaller branches.

That's why your slide says **"Building a Family Tree of Data."** The algorithm literally creates a tree showing how data points are related and grouped together.